Introduction to Machine Learning

Learning notes for document-loading and RAG practice

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions. Instead of writing every rule manually, developers train a model using examples.

Supervised Learning

Supervised learning uses labelled data. Each training example contains input features and a known target. Classification predicts categories, while regression predicts numeric values. Examples include spam detection and house-price prediction.

Unsupervised Learning

Unsupervised learning works with data that does not have labelled answers. It can discover groups, structures, or patterns. Clustering customers into groups is a common example.

Reinforcement Learning

Reinforcement learning trains an agent to make decisions by interacting with an environment. The agent receives rewards or penalties and learns a strategy that aims to maximize long-term reward.

Typical Workflow

A common machine-learning workflow includes collecting data, cleaning data, exploring patterns, selecting features, splitting data into training and testing sets, training a model, evaluating performance, and improving the model.

Evaluation

Classification models may be evaluated using accuracy, precision, recall, and F1-score. Regression models may be evaluated using mean absolute error, mean squared error, and the R-squared score.